

Min Woo Park, Ph.D.

SHORT BIO. My research focuses on causal decision making (NeurIPS-25, Preprint, ICLR-26) and causal effect identification (NeurIPS-24). I investigate how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations. In particular, I develop methods for robust decision-making in settings under environmental uncertainty (NeurIPS-25), domain shifts between training and deployment environments (Preprint), and even scenarios involving counterfactual actions (ICLR-26).

EDUCATIONS

Graduate School of Data Science, Seoul National University

Seoul, Republic of Korea

Integrated M.S. and Ph.D. in Data Science

Sep 2022 – Feb 2027

Advisor: Prof. Sanghack Lee.

Dissertation: “Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals”.

Overall GPA: 4.02 /4.3; Ph.D. Qualification Exam partially waived due to outstanding academic performance.

Department of Mathematics and Statistics, Sejong University

Seoul, Republic of Korea

B.S. in Applied statistics and Mathematics (Double major), *Cum Laude*

Mar 2016 – Aug 2022

Advisors: Prof. Ji Eun Lee and Jin Hee Yoon.

(2.5 years military service)

Overall GPA: 4.04/4.5; Major GPA: 4.46 (Applied Statistics), 4.38 (Mathematics).

PUBLICATIONS

* for joint first authorship and † for corresponding author.

- ◇ Min Woo Park and Sanghack Lee†. [Counterfactual Structural Causal Bandits](#). *International Conference on Learning Representations (ICLR)*, 2026.
- ◇ Min Woo Park, Andy Arditi, Elias Bareinboim†, Sanghack Lee†. [Structural Causal Bandits under Markov Equivalence](#). *Neural Information Processing Systems (NeurIPS)*, 2025, Columbia CausalAI Laboratory, Technical Report (R-122)
- ◇ Min Woo Park and Sanghack Lee†. [On Transportability for Structural Causal Bandits](#). *arXiv preprint*, 2025. under review.
- ◇ Yonghan Jung†, Min Woo Park, and Sanghack Lee†. [Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference](#). *Neural Information Processing Systems (NeurIPS)*, 2024.
- ◇ Min Woo Park*, and Ji Eun Lee*†. [Computation of the iterated Aluthge, Duggal, and Mean transforms](#). *Filomat*, 2023.
- ◇ Min Woo Park*, Taehui Yun*, Youngin Jang, Younsuk Yeom, Jonghwan Kim, LG AI Research, and Sanghack Lee†. Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery. under review.
- ◇ Yesong Choe, Yeahoon Kwon, Min Woo Park, and Sanghack Lee†. Canonical Domain Reduction for Partial Counterfactual Identification. under review.
- ◇ Ji Eun Lee*† and Min Woo Park*. Convergence of the iterated mean transforms of a 2×2 matrix. under review.

AWARDS AND HONORS

Graduate Student Instructor (GSI) scholarship | Seoul National University

Spring 2024, Spring 2025, Spring 2026

BK21 scholarship

Fall 2024, Fall 2025

First place merit scholarship | Sejong University

Fall 2020, Fall 2021

Fourth place award | The 9th College of Natural Sciences Academic Conference, Sejong University Nov 2021

- ▷ Mitigating spurious regression in time-series data via Granger causality.

Third place award | The 8th College of Natural Sciences Academic Conference, Sejong University Nov 2020

- ▷ The significance of the Euler's number and its applications.

PROJECTS

Towards More Scalable Score-based Causal Discovery | LG AI Research Sep 2025 – Aug 2026

- ▷ Leading a project on scalable score-based causal discovery algorithms.
(First-author submission under review.)
 - Developed a novel operator enabling score-based causal discovery algorithms to escape local optima.
 - Investigating parallelization strategies for score-based causal discovery algorithms.

Hyundai NGV Data Boot-Campus 2025 | Hyundai Motor July 2025 – Sep 2025

- ▷ Developed high-voltage battery performance prediction technology.

Hyundai NGV Data Boot-Campus 2024 | Hyundai Motor July 2024 – Sep 2024

- ▷ Developed an intelligent system for predicting vulnerable regions of automotive body panel stiffness using differential geometrybased automatic curvature analysis.

Deep Reinforcement Learning for Recommendation Systems | Coursework Project Mar 2024 – Jun 2024

- ▷ Developed a recommendation system based on DDPG, leveraging multi-modal item features (video, audio, and text) to improve recommendation performance.
 - Source code available at github.com/MinwooPark96/ddpgRLRS.

Optimal Mixed Policies Completeness | Oxford University Aug 2023 – Feb 2024

- ▷ Studied complete characterization of conditions under which the intervention effect of a variable is positive.
Acknowledged for helpful comments in [Toward a Complete Criterion for Value of Information in Insoluble Decision Problems](#). *Transactions on Machine Learning Research* (TMLR), 2024.

TEACHING ASSISTENTS

Causal Inference for Data Science Spring 2024, Fall 2024, Spring 2025, Spring 2026

- ▷ Topics included Bayesian networks and graphical causal inference.

Machine Learning and Deep Learning for Data Science II Fall 2023

- ▷ Covered kernel methods, Gaussian processes, and related topics in machine learning.

Basic Mathematics for Artificial Intelligence (Undergraduate) Spring 2022

- ▷ Covered linear algebra, basic statistics, and probability theory.

ACADEMIC SERVICES, PROFESSIONAL TALKS AND POSTERS

Counterfactual Structural Causal Bandits

ICLR 2026 poster session | Rio de Janeiro, Brazil Apr 2026 (scheduled)

Structural Causal Bandits under Markov Equivalence

ExploreCSR poster session | Seoul, Republic of Korea Dec 2025

NeurIPS 2025 poster session | San Diego, USA Dec 2025

JKAIA 2025, NeurIPS preview session | Seoul, Republic of Korea Nov 2025

SNU GSDS student research seminar | Seoul, Republic of Korea Apr 2025

Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference

NeurIPS 2024 poster session | Vancouver, Canada Dec 2024

SNU GSDS student research seminar | Seoul, Republic of Korea Dec 2024

Conference Reviewer: 2026 NeurIPS, UAI, ICLR, ICML, 2025 ICML

REFERENCE

Sanghack Lee | Graduate School of Data Science, Seoul National University
Associate professor

sanghack@snu.ac.kr

Ji Eun Lee | Department of Mathematics and Statistics, Sejong University
Associate Professor

jieunlee7@sejong.ac.kr

Jin Hee Yoon | Dept. of AI and Information Technology, Sejong University
Associate Professor

jin9135@sejong.ac.kr

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